

GUGA, Russian Federation

WMO: 314210

Lat: 52.700N

Lon: 137.533E

Elev: 197

StdP: 14.59

Time Zone: 10.00 (E10)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.7	(c) -37.9	(d) -46.2	(e) 0.4	(f) -41.2	(g) -42.4	(h) 0.5	(i) -37.5	(j) 9.2	(k) 1.1	(l) 7.6	(m) -0.8	(n) 0.0	(o) N/A	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.2	(c) 87.7	(d) 68.9	(e) 84.1	(f) 67.1	(g) 80.2	(h) 65.4	(i) 71.1	(j) 82.7	(k) 69.2	(l) 80.6	(m) 67.2	(n) 77.3	(o) 4.2	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.7	76.2	65.3	94.3	73.7	63.3	87.6	71.9	35.2	82.7	33.6	81.3	31.9	78.1	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.1	(c) 7.5	(e) -45.9	(f) 94.2	(g) 4.5	(h) 4.3	(i) -49.1	(j) 97.3	(k) -51.7	(l) 99.9	(m) -54.3	(n) 102.3	(o) -57.5	(p) 105.4		
(4) 11.3	(5) 9.1	(6) 7.5	(7) -45.2	(8) 77.1	(9) 4.4	(10) 4.9	(11) -48.3	(12) 80.6	(13) -50.9	(14) 83.5	(15) -53.3	(16) 86.2	(17) -56.5	(18) 89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	-16.3	-3.8	11.7	32.9	46.2	58.5	65.4	61.9	49.7	32.3	6.9	-11.9		
	DBStd	29.73	10.79	10.87	10.74	7.78	7.84	7.77	6.49	5.22	6.77	8.64	11.48	11.32		
	HDD50	9300	2055	1507	1187	514	172	17	0	87	549	1292	1920			
	HDD65	13671	2520	1927	1652	964	582	221	78	125	461	1014	1742	2385		
	CDD50	1253	0	0	0	1	55	272	478	370	77	0	0	0		
	CDD65	147	0	0	0	0	0	26	91	30	0	0	0	0		
	CDH74	2394	0	0	0	0	88	653	1214	428	11	0	0	0		
	CDH80	774	0	0	0	0	13	217	431	112	1	0	0	0		
	WSAvg	1.9	0.9	1.3	1.9	3.1	3.1	2.8	2.3	1.9	1.9	2.0	1.5	0.8		
	PrecAvg	28.5	1.9	1.1	1.5	1.5	2.5	2.4	2.9	3.5	3.0	3.0	2.8	2.3		
(15) Precipitation	PrecMax	37.4	3.9	2.8	3.6	3.6	4.9	5.0	5.8	7.4	5.5	6.3	5.9	4.5		
	PrecMin	20.4	0.0	0.0	0.2	0.7	0.2	0.5	0.3	0.7	1.0	0.9	1.0	0.2		
	PrecStd	3.9	1.1	0.8	1.0	0.6	1.1	1.0	1.5	1.5	1.1	1.3	1.3	1.3		
	PrecStd	3.9	1.1	0.8	1.0	0.6	1.1	1.0	1.5	1.5	1.1	1.3	1.3	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	27.7	49.4	64.8	81.6	91.3	91.5	89.6	75.4	61.3	36.7	27.2		
		MCWB	22.1	23.6	37.9	47.8	59.5	70.1	69.6	70.2	63.2	48.0	32.3	24.6		
	2%	DB	15.7	23.2	41.6	57.6	75.9	86.6	88.4	82.9	70.4	56.5	32.6	17.6		
		MCWB	13.9	20.3	32.5	43.2	56.7	67.3	68.9	67.4	59.5	45.3	28.9	15.6		
	5%	DB	10.9	19.5	36.0	52.6	69.9	82.2	85.3	78.8	65.5	50.5	29.6	12.5		
		MCWB	9.3	16.8	29.6	40.5	54.0	64.0	68.4	65.3	57.3	41.7	26.2	11.1		
	10%	DB	1.0	14.6	31.3	47.9	63.7	77.2	81.7	74.8	61.7	46.2	26.0	7.0		
		MCWB	-0.1	12.1	26.2	37.6	51.0	62.3	67.8	64.2	55.1	40.2	23.2	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	24.0	38.6	48.4	61.3	72.8	74.3	73.7	64.6	49.4	34.0	24.6		
		MCDB	23.9	27.1	47.0	62.9	76.9	87.3	83.5	84.5	72.6	57.9	35.3	27.0		
	2%	WB	14.3	20.7	33.2	44.1	57.9	69.2	71.8	69.8	61.5	45.8	29.7	15.9		
		MCDB	15.8	23.2	40.5	55.9	72.9	84.3	82.6	80.2	67.3	53.4	32.3	17.5		
	5%	WB	9.2	16.9	29.8	41.1	55.1	66.0	70.0	67.5	59.1	43.3	26.1	11.1		
		MCDB	10.9	19.5	35.8	50.7	68.2	78.5	81.4	75.3	64.1	50.1	29.2	12.4		
	10%	WB	-0.1	12.2	26.3	38.3	51.6	63.2	68.4	65.3	56.5	40.2	23.2	5.7		
		MCDB	0.9	14.5	30.9	46.8	62.4	74.8	79.2	72.2	60.8	45.9	26.0	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.8	31.0	32.2	23.3	24.9	27.5	24.2	22.3	21.5	21.9	24.2	22.9		
		MCDBR	20.2	30.9	34.5	33.1	38.7	38.0	32.9	32.7	28.8	29.9	22.6	20.4		
		MCWBR	18.3	27.0	25.7	19.7	20.8	19.4	15.5	17.8	17.2	18.9	18.2	18.6		
	5% WB	MCDBR	19.2	28.1	31.4	31.2	34.7	31.9	27.2	26.5	23.9	27.9	19.6	18.8		
		MCWBR	17.4	24.4	22.9	18.8	19.0	17.4	14.1	15.8	14.7	18.2	16.3	17.5		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.308	0.379	0.494	0.570	0.537	0.551	0.458	0.388	0.390	0.326	0.278		
	taud		2.066	2.009	1.852	1.701	1.732	1.907	1.908	2.183	2.322	2.106	2.043	2.025		
	Ebn at Noon		217	245	246	231	221	232	225	243	248	211	188	187		
	Edh at Noon		24	35	50	67	69	58	57	41	32	32	25	21		
(44) All-Sky Solar Radiation	RadAvg		414	787	1266	1592	1595	1785	1590	1353	1069	664	405	310		
	RadStd		21	32	76	97	166	168	179	148	92	56	34	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page